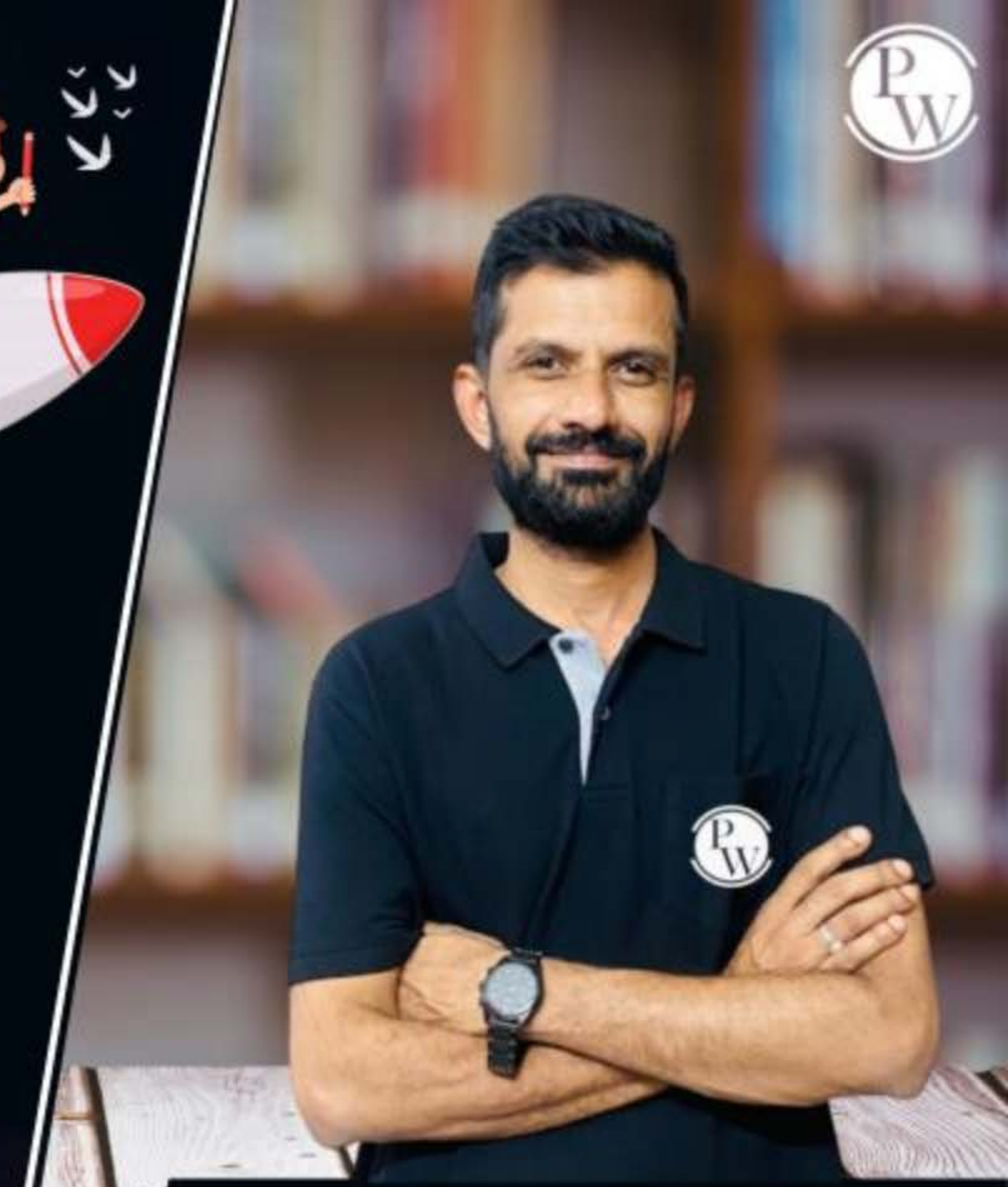
# GATE
## ALL BRANCHES

## General Aptitude

**QUANTITATIVE APTITUDE**

Lecture No.- 11



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# Recap of Previous Lecture

**Topic** Time & Work

# Topics to be Covered

**Topic-1** More on Time and Work

**Topic-2** Pipes and Cistern

[MCQ]

PYQ

#Q. Four persons A, B, C and D are to be seated in a row. C should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is?

Assignment

A 6

B 9

C 18

D 24

$\cancel{C}$

$\boxed{3} \times \boxed{3} \times \boxed{2} \times \boxed{1}$

A/B/C/D (4-1)

A OR B OR D

(4-2)

(4-3)

= 18

[MCQ]

PYQ

#Q. Five different book (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is?

Assignment

**A** 12

**B** 120

**C** 6

**D** 2

$\boxed{3} \times \boxed{2} \times \boxed{1} \times \boxed{S}\ \boxed{R}$

P/Q/T $\quad$ P/Q/T $\quad 1 \times 1$

$(3-1)$

$\underline{6}$

[MCQ]

#Q. A license plate begins with 3 letters. If the possible letters are A, B, C, D and E, how many different ways these letters can be written if no letter is used more than once?

Assignment

$= 60$

$\boxed{5} \times \boxed{4} \times \boxed{3}$

A/B/C/D/E (5-1) (5-2)

A 720

B 120

C 60

D 30

[MCQ]

$720 + 720 + 720 + 720 + 720 + 720$

#Q. In how many different ways can the letters of the word 'PLEADING' be arranged in such a way that the vowels always come together?

Assignment

$V \Rightarrow E, A, I$

$C \Rightarrow P, L, D, N, G$

$6 \times 120 = 720$

$3 \times 2 \times 1 \times 5 \times 4 \times 3 \times 2 \times 1$

E/A/I

P/L/D/N/G

$720 \times 6 = 4320$

(A) 720

(B) 4320

(C) 5040

(D) 3600

[MCQ]

B A B → ? 31 days

A B A B A

#Q. A can do a work in 25 days whereas B in 40 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.......so on. Then in how many days the work would be completed?

Assignment

$\boxed{30\frac{5}{8}}$

$A = \frac{1}{25}$ | $B = \frac{1}{40}$

A. $15\ ^5/_8$ days

B. $30\ ^5/_8$ days

C. 31

D. 16

$\frac{1}{25} + \frac{1}{40} = \frac{13}{200} \times 15 \Rightarrow 2 \text{ days} \times 15$

$\frac{5}{200}$ $\frac{1}{40}$ ?

$\frac{195}{200} \Rightarrow 30 \text{ days}$

1 work A = 25

$\frac{1}{40}$ " $A = 25 \times \frac{1}{40}$

$= \frac{5}{8}$

[MCQ] $1-\frac{1}{4}=\frac{3}{4}$ $A=\frac{1}{20}$ | $B=\frac{1}{30}$ | $C=\frac{1}{60}$

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

$\frac{2}{10}$ 2 days | $\frac{3}{4}$ 15 days | 3 days $\frac{3}{60}=\frac{1}{20}$

A&B&C | B&C | C

$A\&B\&C \rightarrow \frac{1}{20}+\frac{1}{30}+\frac{1}{60}$

$=\frac{6}{60}=\frac{1}{10}$

$\boxed{2+15+3}$

$=20$ days

$\frac{2}{10}+\frac{1}{20}=\frac{5}{20}=\frac{1}{4}$

$B\&C \rightarrow \frac{1}{30}+\frac{1}{60}=\frac{3}{60}=\frac{1}{20}$

1 work: B&C = 20

$\frac{3}{4}$ " " $=20\times\frac{3}{4}=15$

[MCQ]

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

'$x$' days

$$\boxed{\frac{x}{60}+\frac{x-3}{30}+\frac{2}{20}=1}$$

$$x+2x-\cancel{6}+\cancel{6}=60$$

$$\Rightarrow \cancel{3}x=\cancel{60}\ 20$$

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$ | $C = \frac{1}{60}$

#Q. A can do a work in 10 days whereas B in 20 days and C in 60 days. They started the work together. But A left at the end of 3rd day and B left at the end of 5th day. Then the remaining work was done by C in how many days?

| 3 days | 2 days | x days 27 |
|---|---|---|
| A&B&C | B&C | C ? |

$27 - 5 = 22$ days

A. 27

B. 35

C. 31

D. 22

$$\boxed{\frac{x}{60} + \frac{5}{20} + \frac{3}{10} = 1}$$

$$x + 15 + 18 = 60$$

$$x = 27$$

[MCQ]

#Q. If 9 MONKEYS EAT 9 BANANAS IN 9 MINUTES, THEN HOW MANY MONKEYS WILL EAT 45 BANANAS IN 45 MINUTES?

$$9 \times \frac{\cancel{45}}{\cancel{9}} \times \frac{\cancel{9}}{\cancel{45}}$$

| Monkey | Banana | Mins |
|---|---|---|
| 9 | 9 | 9 |
| ? | 45 | 45 |

Demand Delivery

1000 800

? 1200

500 x 3 = 1500

$\cancel{1000} \times \frac{\cancel{1200}}{\cancel{800}}$

2

## Chain Rule:

$\uparrow$ 1 $\times \frac{3}{2}$

$\downarrow$ 1 $\times \frac{2}{3}$

| % | No. |
|---|---|
| 42% | $\rightarrow$ $\boxed{98}$ |
| 63% | $\rightarrow$ ? |

$98 \times \frac{63}{42}$

$42\%$ of $x = 98$

$63\%$ of $x = \frac{63}{100} \times 98 \times \frac{100}{42}$ $\Rightarrow$ $x = 98 \times \frac{100}{42}$ ?

[MCQ]

#Q. 12 men can do a work in 15 days working 8 hours a day. In how many days can 9 men do the same work, working 10 hours a day?

| Men | Days | hrs/day |
|---|---|---|
| 12 | 15 | 8 |
| 9 | ? | 10 |

$$\cancel{15}^{5} \times \frac{\cancel{12}^{4}}{\cancel{9}_{3}} \times \frac{\cancel{8}^{4}}{\cancel{10}_{5}}$$

$$= 4 \times 4 = \underline{16 \text{ days}}$$

[MCQ]

#Q. A hostel with 600 students had sufficient food for 210 days. After 30 days, 240 students left the hostel. Now the remaining food will last for how many days?

Extra days

R.F

300

$180 \times \frac{600}{360}$

S Days T.F.

600 210 330

120

| | | |
|---|---|---|
| 600 | 180 | (210-30) |
| 360 | ? | |
| | 300 | |

Extra Days

120 days

[MCQ]

$6am + 6 hours = 12 PM$

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If that clock represents 1 pm the same day, what must be the correct time?

11:50

$60 \times \frac{\cancel{7}^{1}}{\cancel{70}_{10}}$

$= 6 hrs$

C.W

60

?

W.W

70

7 hrs

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If the correct time is 12 pm the same day, what would be the time shown by that clock?

+

1 PM

[MCQ]

#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

Assignment

A 75

B 100

C 50

D None of these

[MCQ]

#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

Assignment

A ₹ 8000

B ₹ 6000

C ₹ 1800

D ₹ 2040

# Pipes & Cistern

(Time & work)

1 day work

# Pipes & Cistern

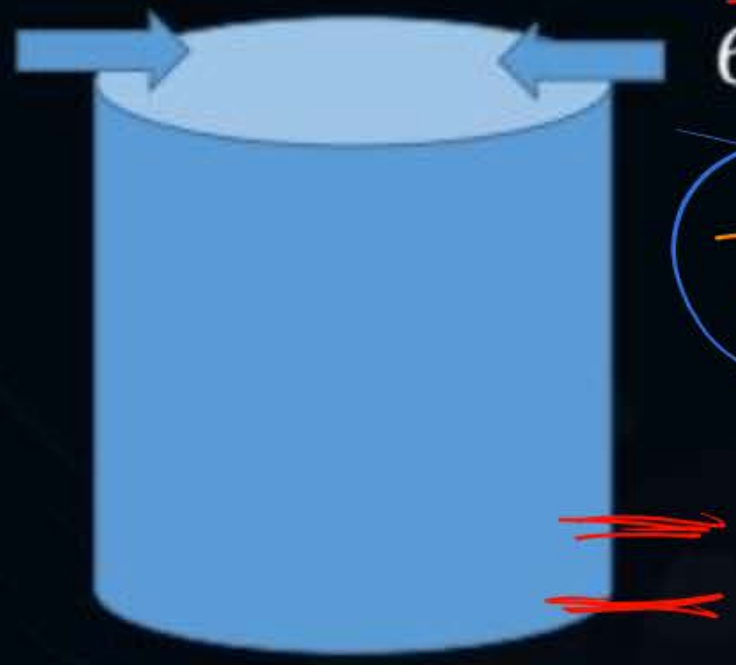
1 hrs work — Empty $(-)$ve

filling $(+)$ve

TAP 'B'

5 hrs

$\frac{1}{5}$



TAP 'A'

6 hrs

$\frac{1}{6}$

TAP C

4 hrs $\quad -\frac{1}{4}$

$$A\&B = \frac{1}{6} + \frac{1}{5} = \frac{11}{30}$$

$$\frac{30}{11} = 2\frac{8}{11} \text{ hrs}$$

$$A\&B\&C = \frac{1}{6} + \frac{1}{5} - \frac{1}{4} = \frac{22-15}{60} = \frac{7}{60}$$

$$\frac{60}{7} = 8\frac{4}{7} \text{ hrs}$$

[MCQ]

$A = \frac{1}{8}$ | $B = \frac{1}{12}$

#Q. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively. Both the pipes are opened together and after 3 minutes, pipe A is turned off. What is the total time required to fill the tank?

A 3 minutes

B 7.5 minutes

C 10 minutes

D 4.5 minutes

$\frac{x}{12} + \frac{3}{8} = 1$

$\Rightarrow 2x + 9 = 24$

$2x = 15$

$\therefore x = \frac{15}{2}$

$= 7.5$ minute

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

A 4.5 hours

B 2 hours

C 6 hours

D 3.2 hours

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

A 30

B 25

C 20

D 15

[NAT]

#Q. A tank consist of three taps A, B and C. Tap A and tap B can fill the tank in 20hrs and 12 hrs respectively, whereas tap C can empty the tank in 18hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A..........so on. In how many hours the tank would be filled?

Brainstorming

# 2 mins Summary

**Topic** Chain Rule Pipes and Cistern

THANK - YOU